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Problem on Ages

→ When ever problem involves only sum or diff of ages, it's a problem of linear equations in one or two variable.

→ Whenever the problem involves product of ages, it's a problem of quadratic eq^n.

→ Age x years ago means ⇒ present age $- x$

→ Age after x years or x years hence means ⇒ present age $+ x$

→ If ratio of present ages is $m:n$, then ages may be assumed as mx & nx.

Q: If twice the son's age is added to father's age the sum is 70. But if twice the father's age is added to son's age, the sum = 95. find age of father & son.

Ans. Son = x
father = y

$$y + 2x = 70$$
$$x + 2y = 95$$

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Q: The sum of ages of father & son is 90 yrs. 15 years ago, father's age was thrice the son's age. find present age?

Ans

$$x + y = 90$$
$$3(x - 15) = (y - 15)$$
$$3x - 45 = y - 15$$
$$3x - y = 30$$

$$3x - 90 + x = 30$$
$$4x = 120$$
$$x = 30 \qquad y = 60$$

formula =

$$\frac{\text{Sum} \pm \text{years (Times} - 1)}{\text{Times} + 1}$$

Result = smaller one present age.

ie.. $\Rightarrow \frac{90 + 15(2)}{4}$

~~father is = Past = true~~

$\Rightarrow \frac{120}{4} = 30.$

Q: The sum of ages of father and son = 60 yrs. 10 years later, father's age was thrice the son's age. find their present ages.

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M-1

$x - y = 60$

$3(x+10) = y+10$

M-2

$\frac{60 - 10(3-1)}{3+1}$

$= \frac{60 - 10 \times 2}{4}$

Son = 10

Father = 50

Q: The ratio of ages of father and son is 17:7. Six years ago this ratio was 3:1. find present age of father.

Ans

$\frac{17x - 6}{7x - 6} = \frac{3x}{x}$

$\cancel{17x^2 - 6x = 21x^2 - 18x}$

$\cancel{12x = 4x^2}$

$17x^2 - 6x = 21x^2 - 18$

$12 = 4x$

$x = 3$

Q: Sum of ages of f & s = 35. Product of age = 150 find age.?

Ans.

$f + s = 35$

$f \times s = 150$

$f + \frac{150}{f} = 35$

$f^2 + 150 = 35f$

$f^2 - 35f + 150 = 0.$

$f^2 - 30f - 5f + 150 = 0$

$f(f-30) - 5(f-30)$

$(f-5)(f-30)$

$f = 5, 30$

$f = 30$

$s = 5$

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Q: I am three times as old as my son. After 5 yrs, I shall be two and a half times as old as my son. How old am I & how old is my son?

Ans $3x = y$ (M-1) Equation

$2.5(x+5) = (y+5)$

$2.5x + 12.5 = y + 5$

$y = 15$, $x = 45$

(M-2) Short Trick

$$\frac{\text{after/before years}\ (\text{times} - 1)}{\text{present time} - \text{will/before time}}$$

$$\frac{5(2.5-1)}{3-2.5}$$

$$\Rightarrow \frac{5 \times 1.5}{0.5}$$

$S \Rightarrow 15$

$F \Rightarrow 45$

Q: Ram is 3 times as old as Shyam. After 15 yrs. Ram will be twice as old as Shyam. find their present ages.

Ans.

$$\frac{15(2-1)}{3-2} \Rightarrow \frac{15}{1} \Rightarrow 15.$$

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Q: A mother is 3 & 1/2 times as old as her daughter. five yrs. ago she was 6 times as old as her daughter, find present ages of mother & daughter.

Ans $\frac{5(6-1)}{6-3.5} \Rightarrow \frac{5\times5}{2.5} \Rightarrow 10$ D

$3.5\times10=35$ M

Q: The age of father 4 years ago was 7 times his son's age then. Presently his age is 4 times the son's age. find present age.

$\frac{4(7-1)}{7-4} \Rightarrow \frac{24}{3}=8$

Q: five years ago, I was twice as old as my son & ten years later, I shall be twice as old as my son. How old are we know?

Ans:

$\frac{5(3-1)+10(2-1)}{(3-2)}$

Q: The ratio of ages of A & B two years ago was 5:3. After 2 years, the ratio of their ages will b/m 7:5. find their present ages?

Ans

$\frac{A-2}{B-2}=\frac{5}{3}$ | $\frac{A+2}{B+2}=\frac{7}{5}$

$3A-5B=-4$ | $5A-7B=4$

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